

Basics Physics-Note Curriculum

One hundred planned prerequisite notes

Thirty standalone textbook drafts plus seventy reserved curriculum scaffolds

Curriculum status. Thirty numbered notes are substantive standalone textbook drafts, version 1.0. Batch 1 comprises Basics 01, 02, 03, 05, 06, 07, 11, 12, 13, and 17. Batch 2 comprises Basics 18, 19, 21, 22, 24, 25, 26, 29, 31, and 32. Batch 3 comprises Basics 34, 35, 36, 37, 39, 40, 41, 42, 44, and 48. The remaining seventy notes are explicitly labeled curriculum scaffolds. Draft status means that derivations, interpretation, a computational laboratory, exercises, symbols, and references are present; it does not imply peer review or final textbook completeness.

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1 How to use this curriculum

The notes form a prerequisite ladder. A student may follow the complete order from Basics 01 through Basics 100 or use the prerequisite tables in advanced module notes. Every basics note remains a separate document; no combined course-volume PDF is maintained. The status printed beside every entry is authoritative for this repository release.

2 Completed standalone batches

Batch 1: mathematical, electromagnetic, and introductory-plasma foundations:

$$\{1, 2, 3, 5, 6, 7, 11, 12, 13, 17\}.$$

Batch 2: plasma collectivity, kinetic moments, ideal MHD, linearization, and elementary Alfvén-wave physics:

$$\{18, 19, 21, 22, 24, 25, 26, 29, 31, 32\}.$$

Batch 3: field-line propagation, Alfvén continua and gap modes, wave-particle resonance, kinetic phase space, guiding-center motion, and energetic-particle distributions:

$$\{34, 35, 36, 37, 39, 40, 41, 42, 44, 48\}.$$

Together, these thirty notes provide a path from mathematical fields and Maxwell equations through ideal-MHD Alfvén waves and into the reduced kinetic physics required by Modules 3–5.

3 I - Mathematical and computational foundations

01. [Basics 01: Units, Dimensions, and Normalization](#)

Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

02. [Basics 02: Vectors, Fields, and Coordinate Systems](#)

Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

03. Basics 03: Vector Calculus for Plasma Physics

Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

04. Basics 04: Ordinary Differential Equations

Curriculum scaffold. Primary connections: All modules, especially Modules 4, 8, 9, and 11.

05. Basics 05: Partial Differential Equations

Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

06. Basics 06: Linear Algebra and Eigenvalue Problems

Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

07. Basics 07: Fourier Analysis and Normal Modes

Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

08. Basics 08: Complex Numbers, Phasors, and Wave Notation

Curriculum scaffold. Primary connections: All modules, especially Modules 4, 8, 9, and 11.

09. Basics 09: Variational Principles and Energy Methods

Curriculum scaffold. Primary connections: All modules, especially Modules 4, 8, 9, and 11.

10. Basics 10: Probability, Statistics, and Monte Carlo

Curriculum scaffold. Primary connections: All modules, especially Modules 4, 8, 9, and 11.

4 II - Electromagnetism foundations

11. Basics 11: Electric and Magnetic Fields

Textbook draft v1. *Primary connections: Modules 1, 3, 4, 5, and 8.*

12. Basics 12: Maxwell Equations

Textbook draft v1. *Primary connections: Modules 1, 3, 4, 5, and 8.*

13. Basics 13: Lorentz Force and Charged-Particle Motion

Textbook draft v1. *Primary connections: Modules 1, 3, 4, 5, and 8.*

14. Basics 14: Electromagnetic Energy and Poynting Theorem

Curriculum scaffold. Primary connections: Modules 1, 3, 4, 5, and 8.

15. Basics 15: Magnetic Pressure, Tension, and Flux

Curriculum scaffold. Primary connections: Modules 1, 3, 4, 5, and 8.

16. Basics 16: Gauge Fields and Electromagnetic Potentials

Curriculum scaffold. Primary connections: Modules 1, 3, 4, 5, and 8.

5 III - Elementary plasma physics

17. [Basics 17: What Is a Plasma?](#)
Textbook draft v1. Primary connections: Modules 1 through 8.
18. [Basics 18: Debye Shielding and Plasma Frequency](#)
Textbook draft v1. Primary connections: Modules 1 through 8.
19. [Basics 19: Plasma Distributions and Temperature](#)
Textbook draft v1. Primary connections: Modules 1 through 8.
20. [Basics 20: Collisions in Plasmas](#)
Curriculum scaffold. Primary connections: Modules 1 through 8.
21. [Basics 21: Single-Fluid and Multifluid Descriptions](#)
Textbook draft v1. Primary connections: Modules 1 through 8.
22. [Basics 22: Moments of the Kinetic Equation](#)
Textbook draft v1. Primary connections: Modules 1 through 8.
23. [Basics 23: Plasma Beta and Dimensionless Parameters](#)
Curriculum scaffold. Primary connections: Modules 1 through 8.

6 IV - Magnetohydrodynamics

24. [Basics 24: Derivation of Ideal Magnetohydrodynamics](#)
Textbook draft v1. Primary connections: Modules 1, 4, 5, and 8.
25. [Basics 25: Conservation Laws in MHD](#)
Textbook draft v1. Primary connections: Modules 1, 4, 5, and 8.
26. [Basics 26: Ideal Ohm's Law and Flux Freezing](#)
Textbook draft v1. Primary connections: Modules 1, 4, 5, and 8.
27. [Basics 27: Resistive and Nonideal MHD](#)
Curriculum scaffold. Primary connections: Modules 1, 4, 5, and 8.
28. [Basics 28: MHD Force Balance and Equilibrium](#)
Curriculum scaffold. Primary connections: Modules 1, 4, 5, and 8.
29. [Basics 29: Linearization of MHD Equations](#)
Textbook draft v1. Primary connections: Modules 1, 4, 5, and 8.
30. [Basics 30: MHD Energy Principle and Stability](#)
Curriculum scaffold. Primary connections: Modules 1, 4, 5, and 8.

7 V - Plasma waves and Alfvén physics

- 31. [Basics 31: Introduction to Waves in Plasmas](#)
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*
- 32. [Basics 32: Alfvén Waves from First Principles](#)
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*
- 33. [Basics 33: Fast and Slow Magnetosonic Waves](#)
Curriculum scaffold. *Primary connections: Modules 4, 5, and 6.*
- 34. [Basics 34: Parallel Wave Number and Field-Line Propagation](#)
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*
- 35. [Basics 35: Alfvén Continuum](#)
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*
- 36. [Basics 36: Coupled Harmonics and Avoided Crossings](#)
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*
- 37. [Basics 37: Toroidal Alfvén Eigenmodes](#)
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*
- 38. [Basics 38: Stellarator Alfvén Eigenmodes](#)
Curriculum scaffold. *Primary connections: Modules 4, 5, and 6.*
- 39. [Basics 39: Wave-Particle Resonance](#)
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*
- 40. [Basics 40: Growth, Damping, and Complex Frequency](#)
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*

8 VI - Kinetic theory and energetic particles

- 41. [Basics 41: Phase Space and Distribution Functions](#)
Textbook draft v1. *Primary connections: Modules 2, 3, 5, and 6.*
- 42. [Basics 42: Vlasov Equation](#)
Textbook draft v1. *Primary connections: Modules 2, 3, 5, and 6.*
- 43. [Basics 43: Boltzmann and Fokker-Planck Equations](#)
Curriculum scaffold. *Primary connections: Modules 2, 3, 5, and 6.*
- 44. [Basics 44: Guiding-Center Theory](#)
Textbook draft v1. *Primary connections: Modules 2, 3, 5, and 6.*

- 45. [Basics 45: Particle Drifts in Nonuniform Fields](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, and 6.
- 46. [Basics 46: Trapped, Passing, and Banana Orbits](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, and 6.
- 47. [Basics 47: Canonical Momentum and Orbit Invariants](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, and 6.
- 48. [Basics 48: Energetic-Particle Distributions](#)
Textbook draft v1. *Primary connections: Modules 2, 3, 5, and 6.*
- 49. [Basics 49: Kinetic MHD and Hybrid Models](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, and 6.
- 50. [Basics 50: Delta-f Particle Methods](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, and 6.
- 51. [Basics 51: Quasilinear Transport](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, and 6.

9 VII - Fusion and burning-plasma foundations

- 52. [Basics 52: Nuclear Fusion Reactions](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, 6, and 7.
- 53. [Basics 53: Fusion Reactivity and Reaction Rates](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, 6, and 7.
- 54. [Basics 54: Alpha-Particle Birth and Slowing Down](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, 6, and 7.
- 55. [Basics 55: Alpha Heating and Burning-Plasma Feedback](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, 6, and 7.
- 56. [Basics 56: Lawson Criterion and Fusion Power Balance](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, 6, and 7.
- 57. [Basics 57: Neutral Beam Injection](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, 6, and 7.
- 58. [Basics 58: Fast-Ion Losses and Wall Heat Loads](#)
Curriculum scaffold. Primary connections: Modules 2, 3, 5, 6, and 7.

10 VIII - Stellarator and magnetic-confinement foundations

- 59. [Basics 59: Magnetic-Confinement Concepts](#)
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.
- 60. [Basics 60: Tokamaks and Stellarators](#)
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.
- 61. [Basics 61: Toroidal Coordinates and Magnetic Surfaces](#)
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.
- 62. [Basics 62: Rotational Transform, Magnetic Shear, and Resonances](#)
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.
- 63. [Basics 63: Stellarator Symmetry and Fourier Geometry](#)
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.
- 64. [Basics 64: Quasisymmetry and Neoclassical Confinement](#)
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.
- 65. [Basics 65: Magnetic-Field Coils and the Biot-Savart Law](#)
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.
- 66. [Basics 66: MHD Equilibrium Codes and VMEC Concepts](#)
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.

11 IX - Computational physics methods

- 67. [Basics 67: Discretization and Numerical Error](#)
Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.
- 68. [Basics 68: Finite-Difference Methods](#)
Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.
- 69. [Basics 69: Finite-Volume Methods](#)
Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.
- 70. [Basics 70: Finite-Element Methods](#)
Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

71. Basics 71: Spectral and Fourier Methods

Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

72. Basics 72: Time-Integration Methods

Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

73. Basics 73: Numerical Eigenvalue Solvers

Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

74. Basics 74: Particle Pushers

Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

75. Basics 75: Particle-in-Cell Methods

Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

76. Basics 76: Interpolation and Conservative Remapping

Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

77. Basics 77: Verification, Validation, and Benchmarking

Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

78. Basics 78: Uncertainty Quantification

Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

79. Basics 79: High-Performance Computing for Simulation

Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

12 X - Multiphysics coupling

80. Basics 80: Multiphysics Modeling

Curriculum scaffold. Primary connections: Modules 8 and 11.

81. Basics 81: Explicit, Implicit, and Partitioned Coupling

Curriculum scaffold. Primary connections: Modules 8 and 11.

82. Basics 82: Fixed-Point and Newton Coupling Methods

Curriculum scaffold. Primary connections: Modules 8 and 11.

- 83. [Basics 83: Multirate and Multiscale Time Integration](#)
Curriculum scaffold. Primary connections: Modules 8 and 11.
- 84. [Basics 84: Conservation Across Module Interfaces](#)
Curriculum scaffold. Primary connections: Modules 8 and 11.
- 85. [Basics 85: Dependency Graphs and Coupling Sequence](#)
Curriculum scaffold. Primary connections: Modules 8 and 11.
- 86. [Basics 86: Digital Twins for Physical Systems](#)
Curriculum scaffold. Primary connections: Modules 8 and 11.

13 XI - Physics-informed machine learning

- 87. [Basics 87: Neural Networks from First Principles](#)
Curriculum scaffold. Primary connections: Modules 9 and 10.
- 88. [Basics 88: Automatic Differentiation](#)
Curriculum scaffold. Primary connections: Modules 9 and 10.
- 89. [Basics 89: Physics-Informed Neural Networks](#)
Curriculum scaffold. Primary connections: Modules 9 and 10.
- 90. [Basics 90: PINN Loss Design and Training Pathologies](#)
Curriculum scaffold. Primary connections: Modules 9 and 10.
- 91. [Basics 91: Eigenvalue PINNs](#)
Curriculum scaffold. Primary connections: Modules 9 and 10.
- 92. [Basics 92: Operator Learning](#)
Curriculum scaffold. Primary connections: Modules 9 and 10.
- 93. [Basics 93: Hamiltonian and Symplectic Neural Networks](#)
Curriculum scaffold. Primary connections: Modules 9 and 10.
- 94. [Basics 94: Surrogate Models for Plasma Simulation](#)
Curriculum scaffold. Primary connections: Modules 9 and 10.
- 95. [Basics 95: Inverse Problems and Data Assimilation](#)
Curriculum scaffold. Primary connections: Modules 9 and 10.

14 XII - Graph neural networks and scientific graphs

- 96. [Basics 96: Graph Theory for Physical Systems](#)
Curriculum scaffold. Primary connections: Module 11, with connections to Module 8.

- 97.** [Basics 97: Graph Neural Networks](#)
Curriculum scaffold. Primary connections: Module 11, with connections to Module 8.
- 98.** [Basics 98: Physical Dependency-Graph Learning](#)
Curriculum scaffold. Primary connections: Module 11, with connections to Module 8.
- 99.** [Basics 99: GNNs for Multiphysics Coupling](#)
Curriculum scaffold. Primary connections: Module 11, with connections to Module 8.
- 100.** [Basics 100: Interpretable and Physics-Constrained Graph Learning](#)
Curriculum scaffold. Primary connections: Module 11, with connections to Module 8.